Real-Time Cyber Threat & URL Classification

Complete Machine Learning Project Expo & Viva Guide

GitHub Repository: github.com/navmi649/cyber-threat-url-classification

1. The 30-Second Teacher Pitch (What to Say First)

"Good morning, Professors. Our project is a Multi-Class Supervised Machine Learning system that inspects raw URL strings in real-time to detect cyber threats. Instead of relying on static blacklists, we engineered a hybrid feature fusion pipeline combining structural lexical attributes with character-level TF-IDF n-grams, driving a Multinomial Logistic Regression model that outputs precise threat probabilities in under 2 milliseconds."

2. The 5 Core Machine Learning Pipeline Steps

* **ML Task Type:** Supervised Learning -> Multi-Class Classification (3 classes: Safe, Malicious, Phishing).
* **Input Vector (X):** Raw URL string (e.g., http://paypal-secure-login.com/login.php).
* **Feature Extraction 1 (Lexical):** 4 structural numbers: URL Length, Dot Count, Hyphen Count, Digit Ratio.
* **Feature Extraction 2 (TF-IDF):** Character-level TF-IDF (3 to 5 n-grams) capturing sub-word attack signatures (.exe, gate, login, verify) invariant to token boundaries.
* **Feature Fusion (scipy):** hstack fuses the dense 4-dim lexical array with the sparse 6,000-dim TF-IDF matrix into a unified 6,004-dim Compressed Sparse Row (CSR) matrix.
* **Classifier & Activation:** Multinomial Logistic Regression using Softmax activation: P(Y=k|X) = exp(w_k^T X + b_k) / sum(exp(w_j^T X + b_j)). Outputs strictly sum to 100%.

3. Threat Categories Explained Simply

* **Class 0: SAFE (Green):** Legitimate domains with standard naming and low punctuation entropy (e.g. google.com, github.com/torvalds/linux).
* **Class 1: MALICIOUS (Red):** Malware delivery, C2 botnet gateways, IP addresses with executable payloads (.exe, .dll, .sh, .vbs).
* **Class 2: PHISHING (Orange):** Brand-jacked credential harvesting portals using hyphen spoofing (e.g. paypal-secure-login.com).

4. Live Demo Walkthrough (Screen Actions)

* **Test 1 (Safe):** Click "Legitimate (Google)" -> Green alert banner (Safe, 99.7% confidence).
* **Test 2 (Malicious):** Click "Malicious (Payload Drop)" -> Red alert banner (Flags IPv4 host, digit ratio, and .exe).
* **Test 3 (Phishing):** Click "Phishing (Fake PayPal)" -> Orange alert banner (Flags brand spoofing and hyphens).
* **Show Charts:** Point to the horizontal bar chart showing Softmax probabilities adding up to 100%.
* **Batch Scanner:** Show the Batch Threat Scanner tab for enterprise multi-URL scanning and CSV report download.

5. Top Examiner Questions & Winning Answers

Q: Why use Machine Learning instead of blacklists?

A: Blacklists are reactive and only block URLs after victims are attacked. Attackers generate disposable domains constantly. ML learns structural patterns to proactively catch zero-day attacks.

Q: Why character-level TF-IDF instead of word-level tokenization?

A: URLs do not have standard word spaces. Attackers use obfuscations like "paypa1" or "login-verify". Word tokenizers fail with Out-Of-Vocabulary (OOV) errors. Character n-grams (3 to 5 chars) slice text into sub-words that catch attack tokens regardless of spacing.

Q: Why did you use StandardScaler on lexical features before fusion?

A: URL length is between 30 and 100+, while TF-IDF values are between 0 and 1. Without scaling, unscaled length would overpower the gradient and suppress the character n-grams. StandardScaler standardizes numerical features so all features

have balanced impact.

Q: Why Logistic Regression with Softmax instead of Deep Learning (LSTM/BERT)?

A: In network firewalls and web gateways, latency is critical. Deep Learning takes 50-300ms per sample and needs GPUs. Our convex fusion model runs in under 1.5ms on standard CPU hardware with high precision and low memory footprint.

Q: What was your experimental dataset and evaluation accuracy?

A: We trained on a balanced dataset of 2,250 URLs (750 Safe, 750 Malicious, 750 Phishing) with an 80/20 stratified split (1,800 train / 450 test), achieving 100% precision, recall, and accuracy on the unseen test set.